

Logistic Regression — Practical Work Report

End-to-End Machine Learning Workflow on Breast Cancer Diagnostic Dataset

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97.37%

MODEL ACCURACY

569

TOTAL SAMPLES

30

INPUT FEATURES

1. Introduction & Objective

This practical work focuses on implementing and evaluating a **Logistic Regression** model for binary classification. The objective is to accurately predict whether a tumor is **Malignant (Class 0)** or **Benign (Class 1)** based on 30 diagnostic cell features computed from digitized images of fine needle aspirate (FNA) tests.

2. Libraries & Tooling

The standard Python data science ecosystem was utilized:

- NumPy** : Efficient multi-dimensional numerical calculations and array processing.
- Pandas** : Structuring data into DataFrames, inspection, indexing, and tabular manipulations.
- Matplotlib** : Visualization interface for generating graphical figures.
- Scikit-learn** : Dataset provider, dataset splitting, standardization, Logistic Regression modeling, and quantitative metrics computation.

3. Dataset Loading & Inspection

The dataset was imported directly via Scikit-learn's `load_breast_cancer()` helper function.

Property	Details
Total Samples	569 observations (patients)
Feature Space	30 numerical float64 measurements (e.g., radius, texture, perimeter, area, smoothness, concavity)
Target Variable	Binary Diagnosis (0 = Malignant, 1 = Benign)
Missing Values	0 missing entries across all columns (100% complete)
Class Distribution	Class 1 (Benign): 357 samples (62.7%) Class 0 (Malignant): 212 samples (37.3%)

4. Preprocessing & Dataset Splitting

To ensure unbiased evaluation, the data was separated into feature matrix `X` (shape: 569 × 30) and target vector `y` (shape: 569).

A. Train-Test Partitioning

We applied an 80/20 train-test split using `random_state=42` for reproducibility:

- **Training Set** (`x_train` , `y_train`): 455 samples (80% of data) used to learn model weights.
- **Testing Set** (`x_test` , `y_test`): 114 samples (20% of data) reserved strictly for evaluation.

B. Feature Standardization

Features have drastically different scales (e.g., *mean area* ranges around ~1000, while *mean smoothness* ranges around ~0.1). Logistic Regression optimizes using gradient descent or coordinate descent, which requires standardized inputs to prevent features with large scales from dominating gradient updates.

Standardization Formula: $z = (x - \mu) / \sigma$

`StandardScaler` fits parameters (μ , σ) solely on `x_train` , and transforms both `x_train` and `x_test` to prevent data leakage.

5. Model Training & Inference

A binary Logistic Regression classifier was instantiated and fitted on the standardized training data:

```
model = LogisticRegression()  
model.fit(X_train, y_train)
```

Upon convergence, the model executed inference on the unseen 114 test samples (`x_test`).

6. Model Evaluation & Performance Metrics

The trained classifier achieved outstanding predictive performance on the test set:

A. Global Accuracy

Overall Test Accuracy: 97.37% (111 correct out of 114 predictions)

B. Confusion Matrix Breakdown

The confusion matrix quantifies precise classification behavior on the test set (114 total instances):

N = 114		Predicted Class	
		Predicted 0 (Malignant)	Predicted 1 (Benign)
Actual Class	Actual 0 (Malignant)	41 (True Negative)	2 (False Positive)
	Actual 1 (Benign)	1 (False Negative)	70 (True Positive)

- **True Negatives (41):** Correctly identified Malignant tumors.
- **True Positives (70):** Correctly identified Benign tumors.
- **False Positives (2):** Malignant tumors incorrectly classified as Benign (Type I error).
- **False Negatives (1):** Benign tumors incorrectly classified as Malignant (Type II error).

C. Detailed Classification Report

Class	Precision	Recall	F1-Score	Support
Class 0 (Malignant)	0.98 (98%)	0.95 (95%)	0.96 (96%)	43 samples
Class 1 (Benign)	0.97 (97%)	0.99 (99%)	0.98 (98%)	71 samples
Macro Average	0.97	0.97	0.97	114 samples
Weighted Average	0.97	0.97	0.97	114 samples

7. Summary & Key Conclusions

- 1. **Pipeline Integrity:** Applying `StandardScaler` before model fitting is essential for gradient optimization in Logistic Regression.
- 2. **High Diagnostic Accuracy:** The model achieves a high F1-Score of 0.96 for Malignant tumors and 0.98 for Benign tumors, indicating robust generalization without overfitting.
- 3. **Clinical Reliability:** Only 1 Benign sample was misclassified as Malignant and 2 Malignant as Benign, making Logistic Regression a strong baseline model for medical decision support.